SIMATS ENGINEERING
ASSESSMENT TOOL 2:

Case Study

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA1520

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Dave's Taxonomy: Precision (Level 3) **Total Marks: 30**
Question Count: 3

Code of Conduct

I _____G. Yashmita (192511032) _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Case Study

1. **Case Study 1: Smart Retail Inventory Management System**

A retail chain implements a cloud-based inventory system connected to store devices Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. **Case Study 2: Cloud-Based Document Collaboration System**

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices.
Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

3. **Case Study 3: Online Food Ordering Platform**

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients.
Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform.
Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

1. Case Study 1: Smart Retail Inventory Management System.

Introduction:

A small retail Inventory Management System is a cloud-based solution that helps retail businesses manage inventory across multiple stores. It connects POS systems, dashboards, and store devices to centralized cloud services through APIs. The system enables real time stock tracking, transaction management, data storage, and demand forecasting.

1. System Architecture:

The System consists of four major components:

Client Layer: Includes POS systems, employee devices and dashboards used to record sales, check stock levels and monitor inventory.

Cloud Services: Process API requests, manage transactions and update inventory information across multiple stores.

Cloud Storage: Stores product details, transaction logs, inventory records and analytics datasets securely.

Platform Services: Provide analytics and demand forecasting by analyzing historical sales and purchasing patterns.

2. Data flow and Communication:

When a customer purchases a product, the POS system sends transaction details to the cloud application through APIs. The cloud service processes the transaction and automatically updates the inventory database.

3. Security Mechanisms:

The System uses multiple security measures to protect business data:

VPN: Provides encrypted communication between retail stores and cloud services.

Firewalls: Prevent unauthorized network access and filter suspicious traffic.

API Gateways: Manage API authentication, authorization, and request validation.

Encryption: Protects data during transmission and storage.

Role-Based Access Control: Restricts access based on employee responsibilities

Multi-Factor Authentication: Adds additional security to user accounts.

4. Benefits:

Real-time inventory tracking across multiple stores.

Reduced manual work and operational costs.

Centralized data management and better decision-making.

Improved data availability through cloud backup and recovery.

5. Conclusion:

The Smart Retail Inventory management System demonstrates how cloud computing, APIs, storage and analytics can improve retail operations.

2. Case Study 2: Cloud-Based Document Collaboration System

Introduction:

A Cloud-Based Document Collaboration System is an online platform that allows multiple users to create, edit, share and manage documents in real time. Similar to Google Docs, users can access the platform through web browsers from different locations and devices. Cloud computing enables centralized document

storage, synchronization, version control, and secure collaboration.

1. System Architecture:

The system consists of the following components:

Client Layer: Users access the platform through web browsers on laptops, desktops, tablets, and mobile devices.

Application Layer: Handles document editing, user requests, realtime collaboration and sharing functionalities.

Cloud Storage: Stores documents in distributed cloud storage to ensure scalability, availability and data durability.

Web APIs: Enable communication and synchronization between multiple clients and devices.

Version Control: Maintains document history and allows users to view previous versions or restore earlier changes.

2. Data Flow and Communication:

When a user creates or edits a document, the changes are sent to cloud servers through web APIs. All changes are saved in cloud storage, while version control maintains

the history of modifications.

3. Network Infrastructure:

The System uses distributed cloud infrastructure and global network connectivity to provide low latency and high availability. Content Delivery Networks (CDNs), load balancers, and geographically distributed data centres help improve response times and distribute user requests efficiently.

4. Security Mechanisms:

Encryption: Protects documents during transmission and storage.

Identity Management: Uses authentication and Multi-Factor Authentication (MFA) to verify user identities.

Access Permissions: Allows document owners to assign viewer, commenter, or editor permissions.

Role-Based Access Control: Restricts access based on user roles and responsibilities.

Audit Logs: Track document access, editing activities, and sharing operations.

5. Benefits:

Enables real-time collaboration from anywhere.

Reduces dependency on local storage and manual file sharing.

Supports document recovery through version control.

Offers scalability and global accessibility.

Conclusion:

The cloud-Based Document Collaborati system demonstrates how cloud computing, distributing storage, web APIs, and real-time synchronization enable efficient teamwork. klith version control, global network infrastruct and strong security mechanisms, the system provides a scalable, reliable, and secure environment for document Collaboration.

3. Case Study 3: Online Food Ordering Platform.

Introduction:

An Online food ordering platform is a cloud-based application that allows customers to browse restaurants menus, place orders, make payments, and track deliveries through web browsers and mobile applications. The platform uses RESTful Web APIs to connect frontend clients with backend services hosted on a cloud platform.

1. System Architecture:

Client Layer: Includes web browsers and mobile applications through which customers browse menus, place orders and track deliveries.

Backend Services: Process customer requests, manage orders, handle payments and communicate with restaurant and delivery services.

RESTful Web APIs: Enable communication between frontend clients and backend services using HTTP methods such as GET, POST, PUT and DELETE.

Cloud Storage: Securely stores menu images, invoices, customer information, and order records.

Cloud Infrastructure: Uses virtual machines and managed Kubernetes to run application services efficiently and support scalability.

2. Data Flow and Communication:

When a customer opens the application, the frontend sends a request to the backend through RESTful APIs to retrieve restaurant details and menus. When an order is placed, the backend validates the request, processes the order and stores the transaction details in cloud storage.

3. Cloud Infrastructure and Scalability:

Virtual machines provide computing resources to run backend applications. Load balancing distributes traffic across available service, improving performance and reliability.

4. Security Mechanisms :

Authentication Tokens

HTTPS

Role based Access Control (RBAC)

Data Encryption

API Security.

5. Benefits :

Provides convenient online food ordering

Supports scalability during peak ordering hours.

Enables real-time order processing and status updates.

Offers secure and centralized data storage.

Improves application availability and reliability.

Conclusion :

The Online Food ordering Platform

demonstrates the effective integration of cloud computing, RESTful APIs, cloud storage, virtual machines and managed kubernetes. These technologies enable real time communication, secure data management, scalability and reliable service delivery. Overall, the cloud based architecture provides an efficient and flexible solution for modern online food ordering businesses